

# BTL1 - Threat Intelligence

## 1. Section Introduction, Threat Intelligence

Welcome to the Threat Intelligence domain of Blue Team Level 1! This domain will cover everything from malicious actors to indicators of compromise, global malware campaigns to information sharing. Below are the learning objectives for this domain.

### Learning Objectives

By the end of this domain you will have achieved the following objectives:

- Understand why threat intelligence is used to aid security operations and risk management, by providing enrichment and context to investigations, threats, and vulnerabilities.
- Understand and explain what threat actors and advanced persistent threats are, including cybercriminal groups and state-sponsored hackers.
- Understand the differences and applications of operational intelligence, tactical intelligence, and strategic intelligence.
- Be able to conduct threat exposure checks using indicators of compromise to search the network and assets for any signs of intrusion using known indicators.
- Be able to conduct research into threat actors using OSINT sources to gather intelligence such as; alternative names, tools, and malware used, targeted organizations, motives, and indicators of compromise.

## 2. Threat Intelligence Explained

Threat Intelligence (TI) is information that has been gathered from a range of sources about malicious actors and their activities. It helps organizations understand the threats they face and how to mitigate them.

### Threat Intelligence Lifecycle

1. **Planning & Direction:** Setting the goals and objectives for the intelligence program.
2. **Collection:** Gathering raw data from various sources (OSINT, commercial feeds, internal logs).
3. **Processing:** Sorting, organizing, and normalizing the collected data.
4. **Analysis:** Examining the processed data to identify patterns, trends, and actionable insights.
5. **Dissemination:** Sharing the finished intelligence with relevant stakeholders (e.g., SOC analysts, executives).
6. **Feedback:** Receiving input on the usefulness of the intelligence to improve future cycles.

## 3. Types of Intelligence

Information is obtained, refined, and then made available to personnel to use accordingly. There are four different types of intelligence:

- **SIGINT (Signals Intelligence):** Involves the interception of radio signals and broadcast communications. Includes COMINT (Communications Intelligence) and ELINT (Electronic Intelligence).
- **OSINT (Open-Source Intelligence):** Information gathered from publicly available sources online (social media, public records, forums).
- **HUMINT (Human Intelligence):** Gathered from human sources through interpersonal contact, espionage, or debriefings.
- **GEOINT (Geospatial Intelligence):** Intelligence derived from the exploitation and analysis of imagery and geospatial information (e.g., satellite images).

## 4. Types of Threat Intelligence

- **Strategic Threat Intelligence:** High-level, non-technical information for decision-makers and executives. Helps with budget spending and policy review.
- **Operational Threat Intelligence:** Focuses on studying threat actors, their motivations, and their Tactics, Techniques, and Procedures (TTPs). Highly technical and requires human analysis to track prolonged campaigns.
- **Tactical Threat Intelligence:** Technical intelligence of immediate value, typically shared as Indicators of Compromise (IOCs) such as malicious IP addresses, URLs, domains, and file hashes. Used for immediate blocking or exposure checks.

## 5. Why Threat Intelligence Can be Useful

- **Cyber Threat Context:** Provides a deeper understanding of who is attacking, why, and how, allowing for better prioritization and defense planning.
- **Incident Prioritization:** Helps SOC analysts determine which alerts pose the highest risk based on known threat actor activity targeting similar organizations.
- **Investigation Enrichment:** Enriches raw logs with context (e.g., flagging a specific IP as a known Command and Control server used by an APT).
- **Information Sharing:** Connecting with other organizations to share IOCs and defensive strategies to stop attacks proactively.

## 6. The Future of Threat Intelligence

### CVEs and CVSS Scores

- **CVE (Common Vulnerabilities and Exposures):** A list of publicly disclosed cybersecurity vulnerabilities. Each vulnerability is assigned a unique CVE ID (e.g., CVE-2021-44228).

- **CVSS (Common Vulnerability Scoring System):** A framework for rating the severity of a vulnerability on a scale of 0 to 10 (10 being the most critical).

## **Vulnerability Context & Predictive Prioritization**

Understanding which vulnerabilities are actively being exploited in the wild (Predictive Prioritization) is crucial, rather than just patching based solely on high CVSS scores. Threat intelligence provides this real-world context.

## **7. Further Reading, Threat Intelligence**

Resources for additional reading include curated lists of awesome-threat-intelligence, open-source intelligence tools (awesome-osint), and blogs from major security vendors detailing advanced malware activity.

## **8. Threat Intelligence Glossary**

This section provides access to a downloadable PDF containing all the terms and acronyms used in the Blue Team Level 1 Threat Intelligence domain to assist with studying.

## **9. Section Introduction, Threat Actors**

This section introduces malicious actors, what they are, how they operate, and why they conduct cyberattacks, covering naming conventions, motivations, and common targets.

## **Learning Objectives**

- Understand different malicious actors and the categories they are typically placed into.
- Understand the different naming conventions used, and why they're used.
- Understand the common motivations for cyberattacks and operations.
- Understand the common targets for different malicious actor categories, and how these align with their motivations.

## **10. Common Threat Agents**

### **What are Threats?**

A threat is any circumstance or event with the potential to adversely impact organizational operations, assets, or individuals through an information system via unauthorized access, destruction, disclosure, or modification of information.

### **What are Threat Actors?**

A threat actor (or malicious actor) is an entity that is partially or wholly responsible for an incident that

impacts – or has the potential to impact – an organization's security.

## Actor Categorization

- **Cyber Criminals:** Motivated primarily by financial gain (e.g., ransomware operators, data thieves).
- **Nation-State (APTs):** Government-sponsored groups conducting espionage, sabotage, or IP theft. Highly resourced and sophisticated.
- **Hacktivism:** Hackers driven by political, social, or ideological motives. Common attacks include website defacement and DDoS.
- **Insider Threat:** Current or former employees, contractors, or partners who have inside

## 11. Motivations

Understanding the motives behind cyberattacks helps defenders anticipate threats. Common motivations include:

- **Financial Motives:** Cybercriminals motivated by monetary gain. Activities include ransomware, selling stolen data, and extortion.
- **Political Motives:** Nation-states and hacktivists driven by geopolitical goals, espionage, disruption, or spreading propaganda.
- **Social Motives:** Actors seeking notoriety, peer recognition, or revenge (e.g., insider threats or script kiddies).
- **Unknown Motives:** Incidents where the attacker's ultimate goal remains unclear during the initial investigation.

## 12. Actor Naming Conventions

Different security vendors use distinct naming conventions to track threat groups:

7. **CrowdStrike:** Uses animal-based cryptonyms. Examples: Bear (Russia), Panda (China), Kitten (Iran), Chollima (North Korea), Spider (Cybercrime).
8. **Mandiant/FireEye:** Uses numbered designations. **APT** (Advanced Persistent Threat) for nation-states, **FIN** for financially motivated groups, and **UNC** for uncategorized groups.

## 13. What Are APTs?

Advanced Persistent Threats (APTs) are highly resourced, stealthy, and goal-oriented threat groups, often state-sponsored.

- **Real-World Examples:** APT28 (Fancy Bear - Russia), Cobalt Group (Financial Cybercrime), APT32 (OceanLotus - Vietnam).
- **What Makes Them Special?** They have massive funding, advanced zero-day exploits, and the patience to remain undetected in networks for months or years.
- **Case Study (Cobalt Group):** Targeted financial institutions using highly customized spear-phishing

emails and malware payloads.

## 14. Tools, Techniques, Procedures (TTPs)

TTPs describe *how* threat actors operate. Understanding them is key to proactive defense.

- **MITRE ATT&CK Framework:** Maps hundreds of adversary techniques across various tactics (e.g., Initial Access, Execution, Persistence, Privilege Escalation).
- **Proactive Defense:** Instead of waiting for an attack, defenders use known TTPs to hunt for threats and implement security controls that block specific adversary behaviors.

## 15 & 16. Activities: Threat Actor Research & Section Review

- **Threat Actor Research:** An activity challenging the student to use free OSINT sources (like MITRE and FireEye reports) to gather intelligence on the APT41 group.
- **Section Review:** A quiz assessing knowledge on threat actors, naming conventions, motivations, and TTPs.

## 17. Section Introduction, Operational Intelligence

This section focuses on the daily tasks of a Cyber Threat Intelligence Analyst:

- Understanding Indicators of Compromise (IOCs) and precursors.
- Applying frameworks like Lockheed Martin Cyber Kill Chain and MITRE ATT&CK.
- Understanding attribution challenges and the Pyramid of Pain.

## 18. Precursors Explained

Precursors are signs that an attack may happen in the future. They are often difficult to detect and act upon.

- **Types of Precursors:**
  - **Port Scanning & Fingerprinting:** Attackers probing the network to map services and vulnerabilities (detectable via firewall/WAF logs).
  - **Social Engineering & Reconnaissance:** Gathering intel on employees or physical security.
  - **OSINT & Bulletin Boards:** Threat groups discussing targets on the dark web or public forums.

## 19. Indicators of Compromise (IOCs) Explained

IOCs are forensic artifacts that strongly suggest a network intrusion has occurred.

- **Examples:** Malicious Email Addresses, IP Addresses (C2 servers), Domain Names/URLs, and File Hashes (MD5, SHA256).
- **IOC Formats:**

- **STIX (Structured Threat Information eXpression):** A standardized language for representing threat intelligence.
- **TAXII (Trusted Automated eXchange of Intelligence Information):** The transport mechanism/protocol used to share STIX data between organizations.

## 20. MITRE ATT&CK Framework

A globally accessible knowledge base of adversary tactics and techniques based on real-world observations.

- **ATT&CK vs. Kill Chain:** While the Cyber Kill Chain proposes a rigid, linear sequence of an attack, MITRE ATT&CK provides a more granular, matrix-based view of the specific techniques used at any stage, making it more practical for mapping defenses.

## 21. Lockheed Martin Cyber Kill Chain A framework developed to outline the phases of a cyberattack, helping defenders detect and contain threats at various stages:

- Reconnaissance: Gathering target information using open sources (OSINT) or active scanning.
- Weaponization: Creating a malicious payload (e.g., embedding malware into a Word document).
- Delivery: Transmitting the payload to the target (via phishing emails or web exploits).
- Exploitation: Triggering a vulnerability to successfully execute the malicious code.
- Installation: Establishing a backdoor or persistence mechanism within the system.
- Command and Control (C2): Establishing a communication channel between the compromised system and the attacker's server.
- Actions on Objectives: Accomplishing the attacker's final goal (e.g., data exfiltration, ransomware encryption).

## 22. Attribution and its Limitations The process of identifying who is responsible for a cyberattack:

- Machine Attribution: Identifying the specific device used (via IP or MAC address).
- Human Attribution: Identifying the actual person executing the attack.
- Ultimately Responsible Attribution: Identifying the organization, crime syndicate, or nation-state directing and funding the attack.
- Limitations: Attribution is extremely difficult due to attackers leveraging False Flags, TOR networks, and proxies to obscure their true identity.

## 23. Pyramid of Pain A visual representation showing the level of difficulty an

## attacker faces when defenders block specific indicators:

- Hash Values: Trivial for the attacker to change.
- IP Address: Easy to change.
- Domain Names: Simple to change.
- Network/Host Artifacts: Annoying for the attacker to adapt to.
- Tools: Challenging; requires the attacker to build or source new tools.
- TTPs (Tactics, Techniques, Procedures): Tough; forces the attacker to completely relearn and change their entire methodology.

## 24 & 25. Operational Intelligence Review & Tactical Intelligence Intro

- Marks the completion of Operational Intelligence and introduces Tactical Intelligence.
- Tactical Intelligence focuses on day-to-day operations like conducting threat exposure checks, continuously monitoring watchlists, and feeding IOCs into automated defenses (SIEM/EDR).

## 26. Threat Exposure Checks Explained

- The process where an analyst actively searches internal logs for newly received Indicators of Compromise (IOCs) from external intelligence feeds.
- Example: Taking a malicious IP address provided by an alert and running a historic search in the SIEM over the past 7 days to see if that IP attempted to scan or breach the network.

## 27. Watchlists / IOC Monitoring

- Automating the detection process by creating permanent watchlists within a SIEM or EDR platform.
- These lists continuously monitor network and endpoint activity, generating immediate automated alerts if any known malicious IP or domain is observed, freeing up the analyst's time.

## 28. Public Exposure Checks Explained

- Analyzing publicly available information about the organization and its employees that could be leveraged by attackers:
  - Social Media Monitoring: Identifying targets for social engineering.
  - Image Metadata: Extracting hidden EXIF data (location, device type) from uploaded photos.
  - Leaked Information: Spotting passwords on sticky notes or internal software versions in the background of office photos.
  - Early Warning Signs: Monitoring for disgruntled employees expressing frustration online.

- Brand Abuse & Impersonation: Detecting fake accounts posing as the organization for phishing.
- Data Breach Dumps: Searching the dark web for compromised employee credentials from previous third-party breaches.

## 29. Threat Intelligence Platforms (TIPs)

- Centralized software solutions used to aggregate, correlate, and analyze threat data from multiple feeds (commercial and open-source).
- They automate the distribution of IOCs directly to security controls like firewalls. Examples include ThreatConnect, Anomali, ThreatQ, and MISP.

## 30. Malware Information Sharing Platform (MISP)

- A highly popular, open-source platform used globally for storing and sharing threat intelligence and IOCs.
- It automatically correlates relationships between malware campaigns and can export detection rules directly to NIDS (like Snort or Suricata) and SIEM systems.

## 31. Threat Intelligence Platform: MISP

**Lab Scenario:** In this lab, you will get hands-on with MISP to perform analysis and research of ingested threat feeds. You will be given different tasks, and must answer questions related to them to complete the lab.

## 32. Activity) Section Review, Tactical Intelligence

*(Note: Image 33 was skipped in your numbering)* Congratulations on completing this section of the Threat Intelligence domain! This knowledge review is designed to test what you have learned about tactical threat intelligence. You will need a score of 70% or above to pass.

## 33. Section Introduction, Strategic Intelligence

This section focuses on strategic roles and responsibilities. A typical day in the life of a Cyber Threat Intelligence Analyst focusing on strategic intelligence typically involves collecting and sharing actionable intelligence with partners and the internal security team to provide context. Strategic Analysts tend to focus on the geopolitical activity of hostile nations, working to monitor if there is or will be an increased threat from that nation and their associated Advanced Persistent Threats (APTs). **Learning Objectives:**

- Understand how indicators of compromise and tactics, techniques, and procedures can be shared between organizations and the value it generates.
- Develop your understanding of how threat intelligence teams work to track malicious campaigns and report this information to the security team so they can prepare for attacks.
- Compare the strengths and weaknesses of both OSINT-acquired intelligence and intelligence purchased from vendors.

## 34. Intelligence Sharing and Partnerships

If an organization has an established threat intelligence team, someone will likely be responsible for connecting with other organizations to join or form an **Information Sharing and Analysis Center (ISAC)**. These are typically industry-specific groups comprised of multiple organizations in order to share actionable intelligence such as indicators of compromise, precursors, and information about attacks and threats.

**Example:** The Aviation ISAC (a-isac) is a collective of organizations operating within the aviation industry coming together to share threat intelligence.

- VirusTotal
- URLhaus
- AlienVault Open Threat Exchange (OTX)
- AbuseIPDB

Paid-For Intelligence:

Typically used by large enterprises. Dedicated threat intelligence teams ingest this data to power defenses.

Providers include:

- FireEye
- Recorded Future
- CrowdStrike
- Flashpoint
- Intel 471

## 35. Traffic Light Protocol (TLP)

The TLP is a set of designations used to ensure that sensitive information is shared with the appropriate audience. It relies on a color-coded scheme:

- **TLP:CLEAR:** Information carries minimal risk. Subject to standard copyright rules, it can be freely shared.

- **TLP:GREEN:** Information is useful for the awareness of all participating organizations within the community. Not for public release.
- **TLP:AMBER:** Information requires support to be effectively acted upon. Share only on a strict need-to-know basis within the organization and clients.
- **TLP:AMBER+STRICT:** Restricts sharing strictly to the organization only.
- **TLP:RED:** Information for the eyes and ears of individual recipients only. Highly restricted.

## 36. Permissible Action Protocol (PAP)

Conceptualized in 2016 under MISP. Unlike TLP (which classifies how information is shared), PAP classifies actions based on the potential exposure of sensitive data to malicious entities.

- **PAP:CLEAR:** Actions are relatively unrestricted.
- **PAP:GREEN:** Permits controlled, non-intrusive actions with malicious sources (e.g., blocking incoming threats at the firewall).
- **PAP:AMBER:** Confined to passive data handling, ensuring actions remain undetected by malicious sources.
- **PAP:RED:** Exclusively around detection and investigation. Actions must be invisible to adversaries (e.g., searching non-production environments based on logged data).

## 37. Activity) Section Review, Strategic Intelligence

Congratulations on completing this section of the Threat Intelligence domain! This knowledge review is designed to test what you have learned about strategic threat intelligence. You will need a score of 70% or above to pass.